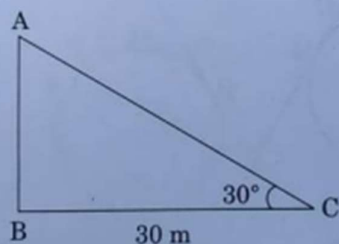


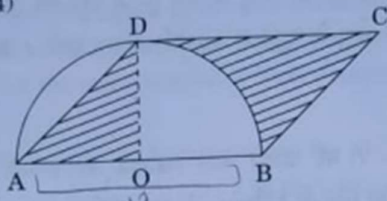
3. The radius of a sphere (in cm) whose volume is  $12\pi \text{ cm}^3$ , is
- (A) 3  
(B)  $3\sqrt{3}$   
(C)  $3^{2/3}$   
(D)  $3^{1/3}$

20. In Figure-4, the angle of elevation of the top of a tower from a point C on the ground, which is 30 m away from the foot of the tower, is  $30^\circ$ . Find the height of the tower.



21. A cone and a cylinder have the same radii but the height of the cone is 3 times that of the cylinder. Find the ratio of their volumes.

30. In Figure-8, ABCD is a parallelogram. A semicircle with centre O and the diameter AB has been drawn and it passes through D. If  $AB = 12 \text{ cm}$  and  $OD \perp AB$ , then find the area of the shaded region. (Use  $\pi = 3.14$ )



37. A statue 1.6 m tall, stands on the top of a pedestal. From a point on the ground, the angle of elevation of the top of the statue is  $60^\circ$  and from the same point the angle of elevation of the top of the pedestal is  $45^\circ$ . Find the height of the pedestal. (Use  $\sqrt{3} = 1.73$ )

39. In a cylindrical vessel of radius 10 cm, containing some water, 9000 small spherical balls are dropped which are completely immersed in water which raises the water level. If each spherical ball is of radius 0.5 cm, then find the rise in the level of water in the vessel.

40. If a line is drawn parallel to one side of a triangle to intersect the other two sides at distinct points, prove that the other two sides are divided in the same ratio.

32. If  $\tan^{-1} \left( \frac{y}{x} \right) = \log \sqrt{x^2 + y^2}$ , prove that  $\frac{dy}{dx} = \frac{x+y}{x-y}$ .

OR

If  $y = e^{a \cos^{-1} x}$ ,  $-1 < x < 1$ , then show that

$$(1-x^2) \frac{d^2y}{dx^2} - x \frac{dy}{dx} - a^2 y = 0$$

55/2/2